

Signals and Systems

$$x(t) \quad T$$

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

$$\omega_0 = \frac{2\pi}{T}$$

$$\mathbf{1. \quad FS \quad \hat{x}[k]}$$

$$x(t) \quad [-T/2, T/2] \quad x(t) = \delta(t)$$

$$\hat{x}[k] = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt$$

$$\delta(t)$$

$$\hat{x}[k] = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \cdot e^0 = \frac{1}{T}$$

$$x(t)$$

$$x(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T} e^{jk\omega_0 t}$$

2. FT

$$e^{j\omega_c t} \longleftrightarrow 2\pi\delta(\omega - \omega_c)$$

$$X(j\omega) = \mathcal{F} \left\{ \sum_{k=-\infty}^{\infty} \frac{1}{T} e^{jk\omega_0 t} \right\} = \frac{1}{T} \sum_{k=-\infty}^{\infty} \mathcal{F} \{ e^{jk\omega_0 t} \}$$

3.

$$X(j\omega) = \sum_{k=-\infty}^{\infty} \frac{2\pi}{T} \delta(\omega - k\omega_0)$$

$$\omega_0 = \frac{2\pi}{T}$$

$$X(j\omega) = \sum_{k=-\infty}^{\infty} \frac{2\pi}{T} \delta\left(\omega - k\frac{2\pi}{T}\right)$$

$$\longleftrightarrow$$

$$\begin{array}{c} \bullet \\ \bullet \end{array} \quad T \quad \frac{2\pi}{T}$$